Ryan Hubbart

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Education

Purdue University — West Lafayette, IN
Bachelor of Science in Computer Science — *Aug. 2024 – May 2028*

Concentration: Systems Software

Experience

Lynco Products — Quad Cities Metro Area
Full Stack Software Engineer Intern — *June 2026 – Aug. 2026*

- Shipped internal React tools to production, including a shipping rate engine that queries carrier APIs such as FedEx and picks the cheapest carrier per order by shipment size and quantity, replacing manual rate comparison.
- Moved untracked production SQL stored procedures into version control, so changes are reviewable and reversible.
- Provisioned an in-house Debian development server for a five-person team with LDAP auth and Samba shares, pinning per-project Node toolchains with Volta so legacy React apps run alongside newer code.

BoilerMake — West Lafayette, IN
Outreach Team Lead — *April 2025 – Present*

- Lead a six-person team that grew engagement 1.5× year-over-year for the 500-person BoilerMake hackathon.
- Analyzed 1,000+ applications for gaps by major, year, and school, then redirected outreach to untapped groups.
- Built Sponsi, a TypeScript Discord bot for sponsor outreach: an LLM orchestrator delegates to narrow agents, each limited to its own MCP tools (Google Sheets, Hunter, web search), with tab locks and approvals enforced in code.

Purdue Grand Prix Foundation — West Lafayette, IN
External Relations Team — *Sept. 2025 – Present*

- Diagnosed gate-entry and ticketing inefficiencies that were costing the foundation third-party fees, then designed and built an in-house alternative for its 3,000-attendee event.

Purdue Hackers — West Lafayette, IN
Workshops Organizer — *Aug. 2026 – Present*

- Pick emerging technologies to spotlight and run hands-on workshops for Purdue's student developer community.

Projects

Event Ticketing System | *Python, FastAPI, SQLAlchemy, SQLite, Stripe, Ed25519* — Oct. 2025 – Present

- Built an end-to-end ticketing platform to replace third-party ticketing for Purdue Grand Prix (3,000 attendees), covering Stripe checkout, emailed QR tickets, and gate scanning.
- Designed gate entry to run with no network: tickets are Ed25519-signed QR codes that standalone door laptops verify with only the public key, reading USB barcode scanners against a local SQLite snapshot.
- Prevented overselling and double entry by putting each check inside a single conditional UPDATE so the database decides every race; Stripe webhooks issue mixed-tier orders all-or-nothing, exactly once.
- Wrote 90 hermetic PyTest tests, including races of 150 concurrent buyers and 25 simultaneous scans of one ticket; CI also proves the Alembic migration builds exactly the schema the models describe.

Swatch | *Python, FastAPI, NumPy, scikit-learn, CLIP, Linux* — Sept. 2026

- In four hours at a hackathon, built a clothing recommender over 1,634 CLIP-embedded items scraped from 14 brands; mean-centering removed the shared product-photo look, cutting mean similarity from 0.73 to 0.04.
- Modeled taste as up to four k-means clusters scored by best match, rather than one averaged vector, since averaging dissimilar preferences ranks a mediocre hybrid above either liked item.
- Balanced exploration against exploitation by round-robining the feed across clusters so the strongest cannot starve the rest, plus active learning on items between two tastes and a random wildcard.

Technical Skills

Languages: Python, C, C++, TypeScript, JavaScript, Java, SQL
Backend: FastAPI, Node.js, REST APIs, SQLAlchemy, PostgreSQL, SQLite, Stripe API
Frontend: React, Svelte, HTML/CSS | **Data**: NumPy, Pandas, scikit-learn
Tools: Linux (Debian), Docker, Git, GitHub Actions CI, PyTest, Alembic, LDAP, Samba